

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I M.Kyathi Sai Priya (192572101) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __M.Kyathi__

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

Ans: Hypervisor:

Definition : A hypervisor (Virtual Machine Monitor - VMM) is software that enables multiple Virtual Machines (VMs) to run on a single physical server.

There are four types of hypervisors:

- Type-1 (Bare-Metal)
- Type-2 (Hosted)
- Type-0
- Hybrid Hypervisor

Comparison of Type-1 and Type-2 Hypervisors for Mission-Critical Cloud Datacenters:

Comparison Table :

Parameter	Type-1 Hypervisor (Bare-Metal)	Type-2 Hypervisor (Hosted)
Installation	Installed directly on physical hardware	Installed on top of a host operating system
Performance	Very high performance with low overhead	Lower performance due to host OS overhead
Security	More secure as there is no host OS	Less secure because attacks on the host OS can affect VMs
Resource Utilization	Efficient use of CPU, memory, and storage	Less efficient because resources are shared with the host OS
Reliability	Highly reliable and stable	Depends on the stability of the host OS
Scalability	Supports a large number of VMs	Suitable for a limited number of VMs
Manageability	Centralized management, live migration, high availability	Basic management features, limited enterprise support
Maintenance	Easier to maintain in enterprise environments	Host OS updates may affect VM operation
Examples	VMware ESXi, Microsoft Hyper-V, Xen	Oracle VM VirtualBox, VMware Workstation
Best Used For	Cloud datacenters, enterprise servers, mission-critical applications	Personal computers, testing, software development

Analyzing :

Performance

- Type-1 hypervisors run directly on hardware, providing faster execution and lower latency.
- Type-2 hypervisors depend on the host operating system, increasing processing overhead.

Security

- Type-1 offers stronger isolation between virtual machines and has a smaller attack surface.
- Type-2 is more vulnerable because attacks on the host operating system can affect all virtual machines.

Manageability

- Type-1 supports centralized administration, live migration, load balancing, and high availability.
- Type-2 provides basic management and is mainly suitable for development or testing.

Justification :

For mission-critical applications, Type-1 Hypervisor is the better choice because it:

- Provides maximum performance.
- Ensures stronger security.
- Offers better VM isolation.
- Supports high availability and disaster recovery.
- Easily scales to support large cloud datacenters.

2. cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Ans: Virtualization Architecture for Workload Isolation

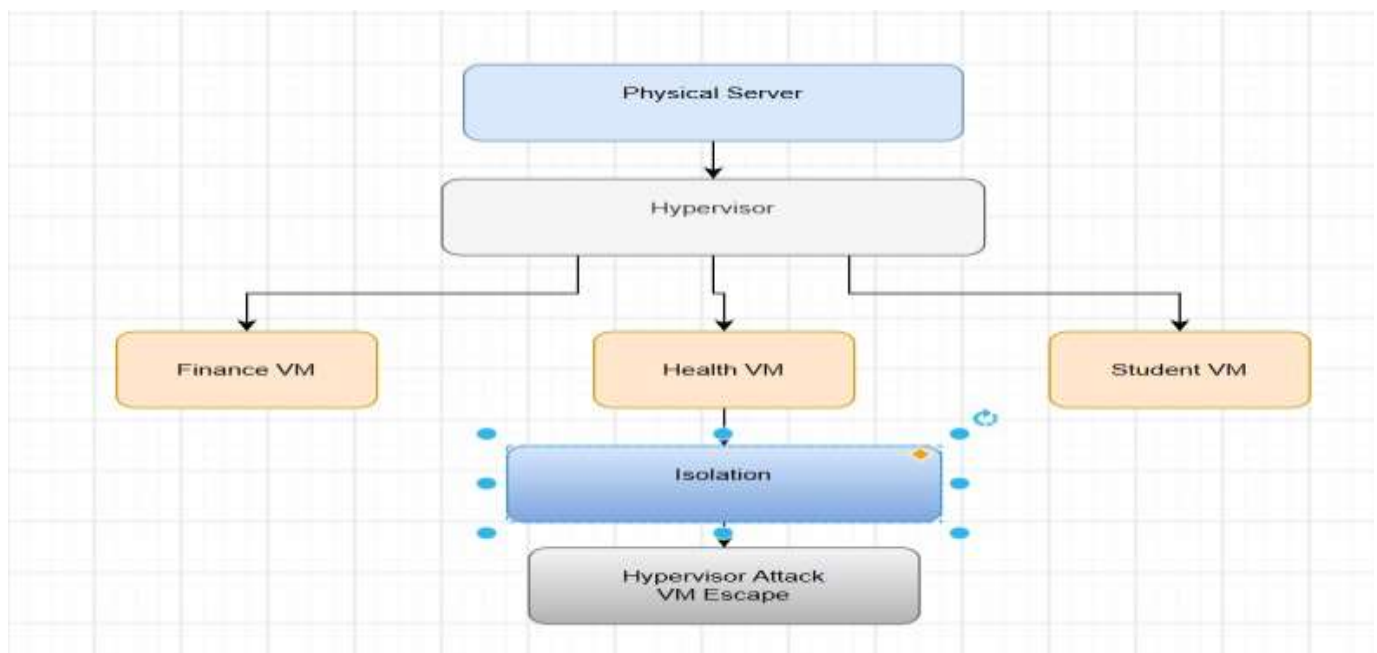
Definition:

Virtualization allows multiple organizations to safely share the same physical server while remaining isolated from each other.

Given : A cloud provider hosts:

- Finance applications
- Healthcare applications
- Student portals

All run on the same hardware.



How Virtualization Supports Isolation :

1. CPU Isolation

- It Separate virtual CPUs for each VM.
- One VM cannot affect another.

2. Memory Isolation

- It dedicated to virtual memory.
- It also Prevents unauthorized access.

3. Storage Isolation

- Separates virtual disks.
- Independent file systems.

4. Network Isolation

- VLANs
- Virtual switches
- Software-defined networking

5. Hypervisor Isolation

- Controls communication between VMs.
- Prevents direct access.

Benefits

- Better security
- Resource sharing
- Reduced cost
- High availability

Major Attack Surfaces :

1. VM Escape

- Attacker escapes from VM to hypervisor.
- May compromise other VMs.

Solution

- Hypervisor patching
- Strong access control

2. Virtual Network Attacks

- Packet sniffing
- ARP spoofing
- VLAN hopping

Solution

- Firewalls
- VLAN segmentation
- Network encryption

Additional Security Measures

- Multi-factor authentication
- Encryption
- IDS/IPS
- Role-Based Access Control

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Ans :

Given data :

CPU Usage = 92%

Memory Usage = 88%

Storage I/O latency = 35 ms

VM Boot time = 170 Sec

Analysis

1. CPU Bottleneck

- CPU utilization is very high.
- Processor becomes overloaded.
- VMs respond slowly.

Corrective Action

- Add CPU cores.
- Balance workloads.
- Migrate VMs.

2. Memory Bottleneck

- Memory usage is 88%.
- Less free RAM.
- Possible memory swapping.

Corrective Action

- Increase RAM.
- Optimize VM memory allocation.
- Remove unused VMs.

3. Storage Bottleneck

- Storage latency is 35 ms.
- Normal latency should be much lower.
- Slow disk access delays applications.

Corrective Action

- Upgrade to SSD/NVMe.
- Use storage tiering.
- Improve storage controller performance.

4. Long VM Boot Time

- 170 seconds is too high.
- Indicates storage or CPU bottleneck.

Corrective Action

- Optimize startup services.
- Improve storage speed.
- Allocate adequate resources.

Likely Bottlenecks

- High CPU utilization
- High memory usage
- Slow storage performance

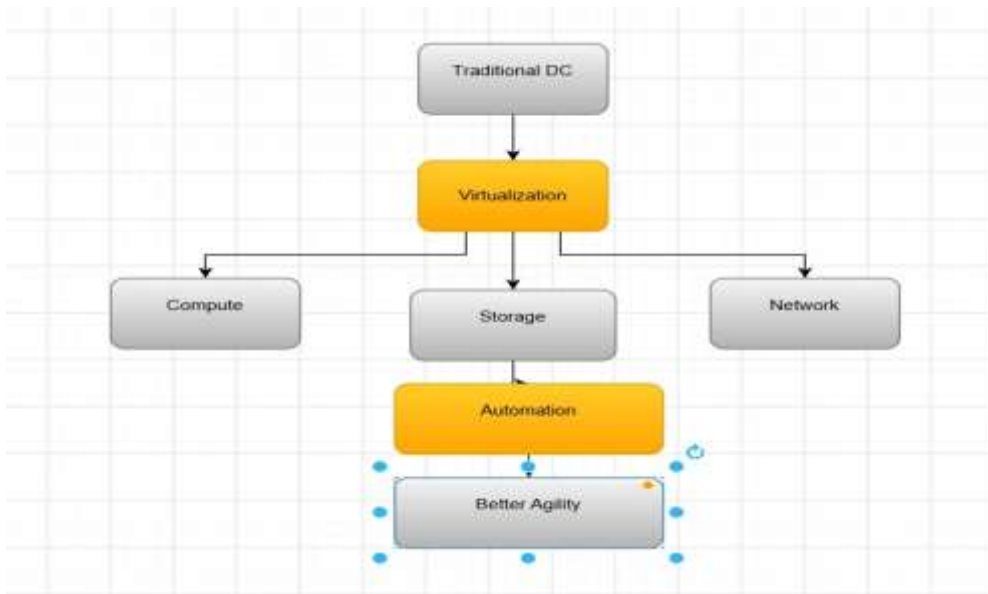
Overall Recommendation :

- Add compute resources
- Upgrade storage
- Enable load balancing
- Monitor resource utilization continuously

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Ans: Software Defined Datacenter(SDDC) vs Traditional Datacenter :

- A Software Defined Datacenter virtualizes all infrastructure resources, while a traditional datacenter depends heavily on dedicated hardware.



Comparison :

Parameter	Traditional Datacenter	Software Defined Datacenter (SDDC)
Architecture	Hardware-based	Software-based
Compute	Physical servers	Virtual machines and virtualized compute
Storage	Dedicated physical storage	Software-defined storage (SDS)
Networking	Physical switches and routers	Software-defined networking (SDN)
Provisioning	Manual configuration	Automated provisioning
Scalability	Limited and slow	Highly scalable and flexible
Resource Utilization	Lower utilization	Higher utilization through virtualization
Management	Manual administration	Centralized software management
Deployment Time	Takes days or weeks	Takes minutes or hours
Cost	High hardware and maintenance cost	Lower operational cost
Disaster Recovery	Manual backup and recovery	Automated backup and disaster recovery
Agility	Low	High

How SDDC Improves Agility :

1. Compute Virtualization

- Creates multiple virtual machines on one physical server.
- Enables quick deployment and resource allocation.

2. Storage Virtualization

- Combines multiple storage devices into a single virtual storage pool.
- Improves storage efficiency and simplifies management.

3. Network Virtualization

- Uses Software Defined Networking (SDN) to create virtual networks.
- Allows faster network configuration and better traffic management.

Advantages of SDDC

- Faster deployment of applications.
- Better scalability.
- Improved resource utilization.
- Reduced operational costs.
- High availability and disaster recovery.
- Centralized management with automation.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Ans : Multi-Cloud Federation Identity Problem :

A multi-cloud federation allows organizations to use services from multiple cloud providers. Identity mismatches can prevent users from accessing resources consistently.

This causes:

- Login failures
- Duplicate user accounts
- Inconsistent permissions
- Security risks

Secure Identity Design :

1. Federated Identity Management

- Single digital identity across all clouds.
- Users authenticate once.

2. Single Sign-On (SSO)

- One login provides access to multiple cloud services.
- Reduces password management.

3. Identity Provider (IdP)

- Central server verifies user identity.
- Cloud providers trust the IdP.

4. Role-Based Access Control (RBAC)

- Users receive permissions based on roles.
- Ensures least-privilege access.

Privacy Protection

- Encrypt identity information.
- Use secure authentication.
- Share only necessary user attributes.
- Maintain audit logs.

Benefits :

- Improved security
- Simplified user management
- Better privacy
- Consistent access across clouds
- Reduced administrative effort